

First-Order RL and RC Filters – Basic Frequency Response Analysis

Due Tuesday 2/13/2024

You are not allowed to work with anyone on this project or any other project or hw. All your work should be yours only. You are not allowed to acquire code on line or from any other source. You need to do the code yourself.

Any violation will result in an automatic failing of the class.

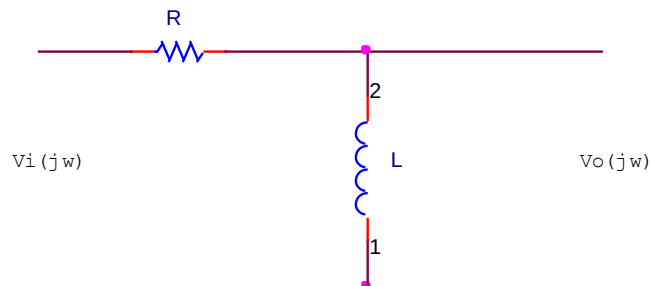
Filters are very useful in selecting a certain band of frequencies or attenuating a certain band of frequencies. Filters are very useful in communication and electronics.

For all the circuits determine the transfer function $H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$ of the circuit using

a phasor analysis approach using the parameters shown below. Generate in MATLAB (a) a semi-logarithmic plot of the circuit gain vs. frequency and phase-shift vs. frequency. These plots are called Bode plots and (b) a polar plot of the circuit gain vs. phase-shift. These plots are called Nyquist plots.

- For all the circuits below determine the transfer function of the circuit using a phasor analysis approach. Using the following circuit parameters, generate in MATLAB:
 - (a) a semi-logarithmic plot of the circuit gain vs. frequency and phase-shift vs. frequency, and
 - (b) a polar plot of the circuit gain and phase-shift.

Example (a): High Pass: $R=100; L=10\text{e-}3$;



Solution of (a)
(a)

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{j\omega L}{R + j\omega L}$$

%Example code:

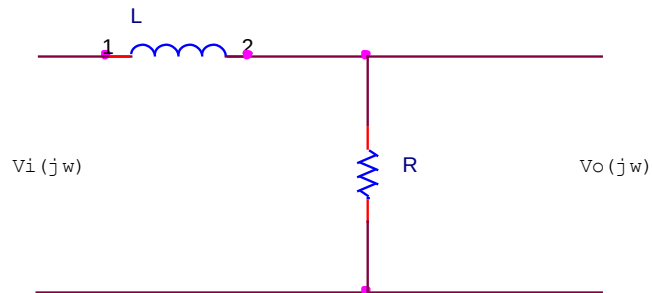
```
R=100;L=10e-3;
f=logspace(1,5,1000);
```

```
% Method 1
num=[L 0];
den=[L R];
jw=j*2*pi*f;
H_jw=polyval(num,jw)./polyval(den,jw);
```

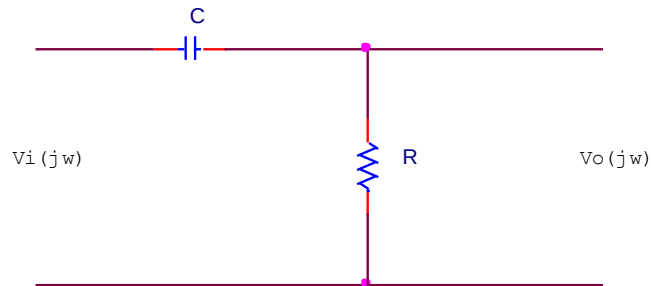
```
% Method 2
w=2*pi*f;
H_jw=j*L*w./(R+j*w*L)
```

```
figure;
subplot(211);semilogx(f,abs(H_jw));grid
title('RL Circuit (a)')
subplot(212);semilogx(f,angle(H_jw)*180/pi);grid
figure;
polar(angle(H_jw),abs(H_jw));grid
title('RL Circuit (a)')
```

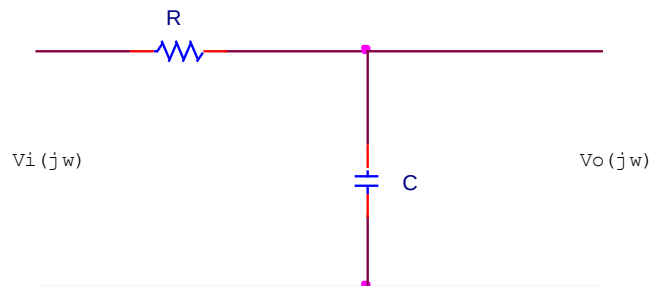
(b) Low pass: $R=100; L=10\text{e-}3;$



(c) High pass: $R=1\text{e}3, C=1\text{e-}6.$

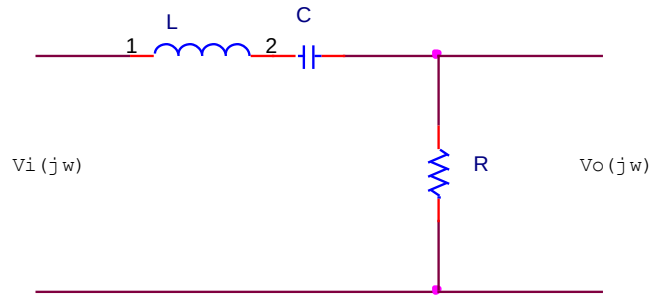


(d) Low pass: $R=1\text{e}3, C=1\text{e-}6.$

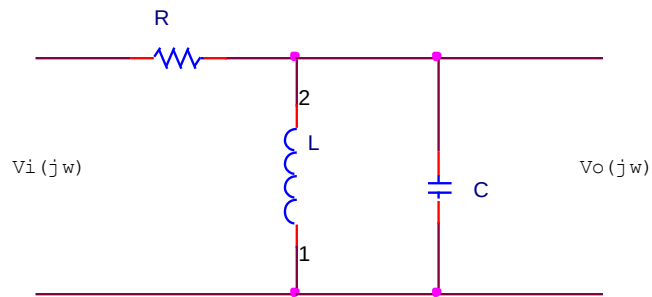


Second-Order RLC Filters – Basic Frequency Response Analysis

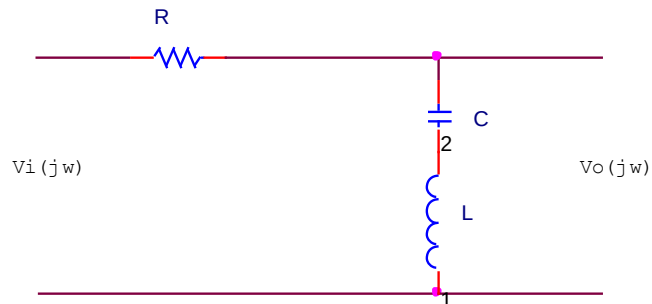
(e) Band pass filter: $R=1\text{k}\Omega, C=1\mu\text{F}, L=10\text{mH}.$



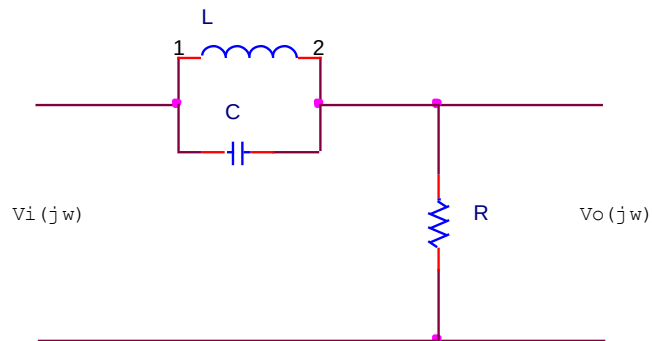
(f) Band pass filter: $R=100$; $L=1\text{e-}3$; $C=1\text{e-}6$;



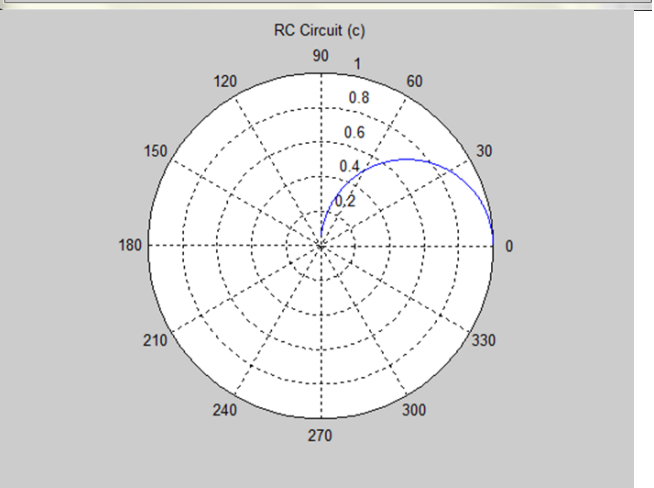
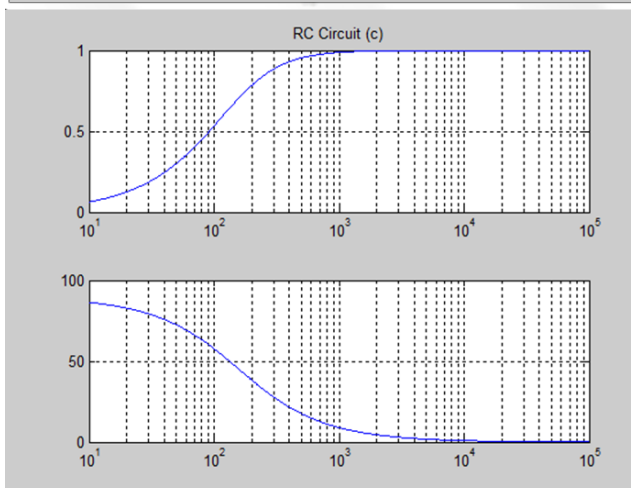
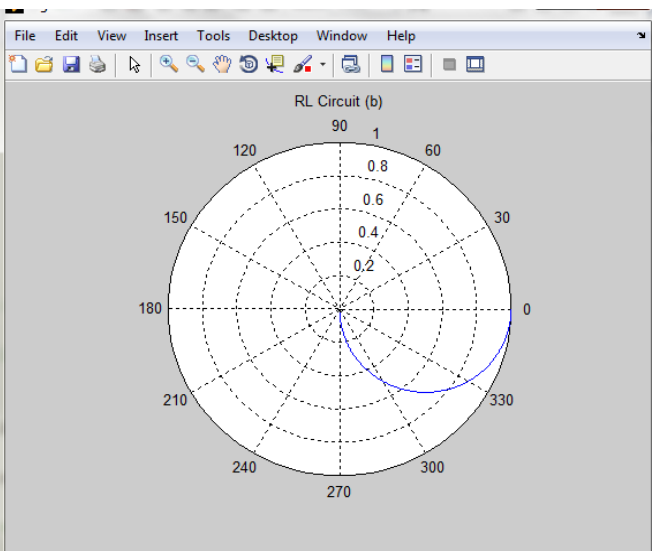
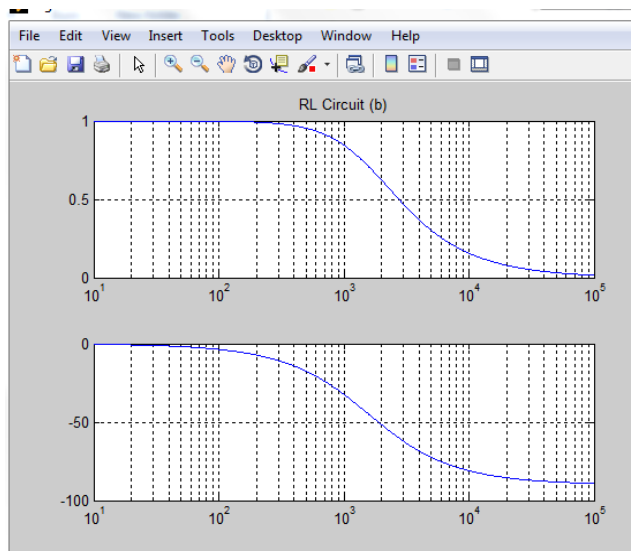
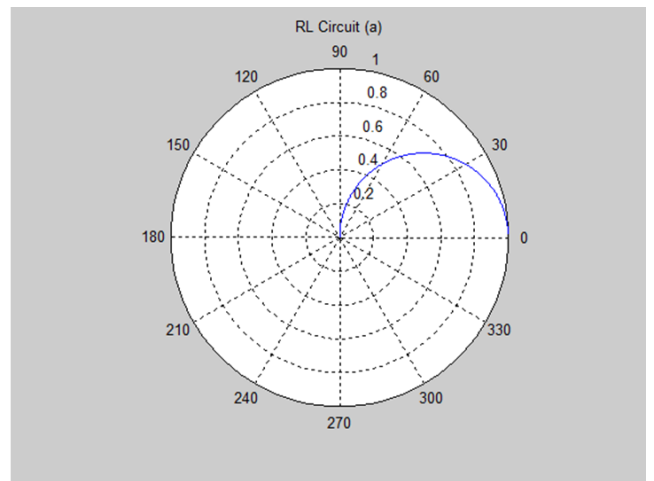
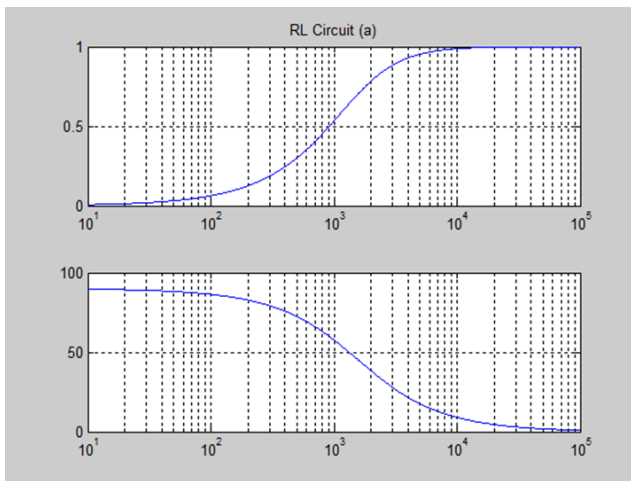
(g) Notch filter: $R=100\Omega$, $L=1\text{mH}$, $C=1\mu\text{F}$.

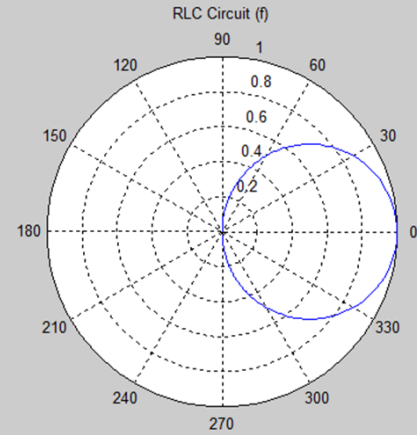
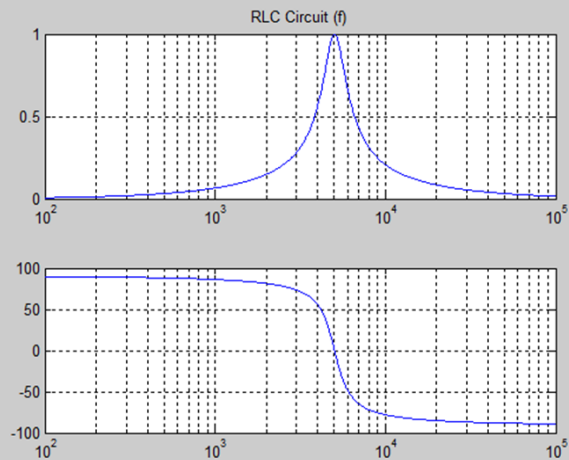
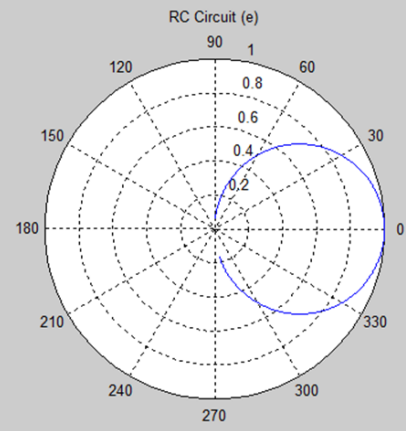
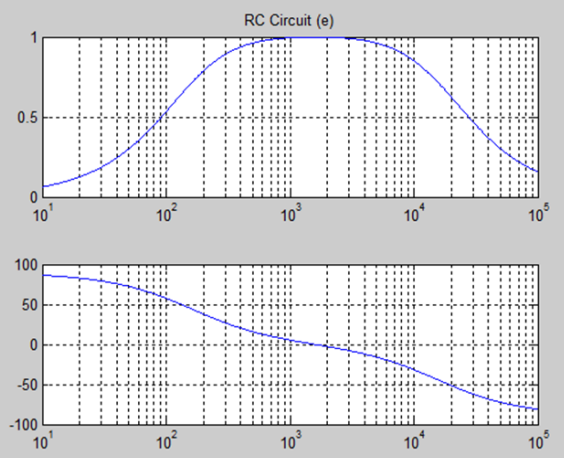
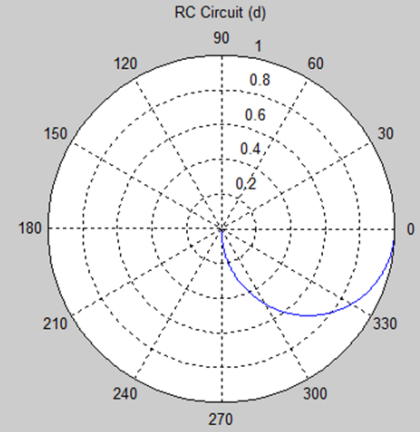
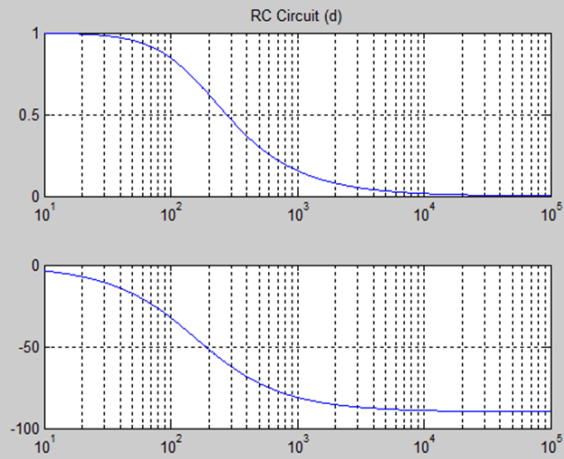


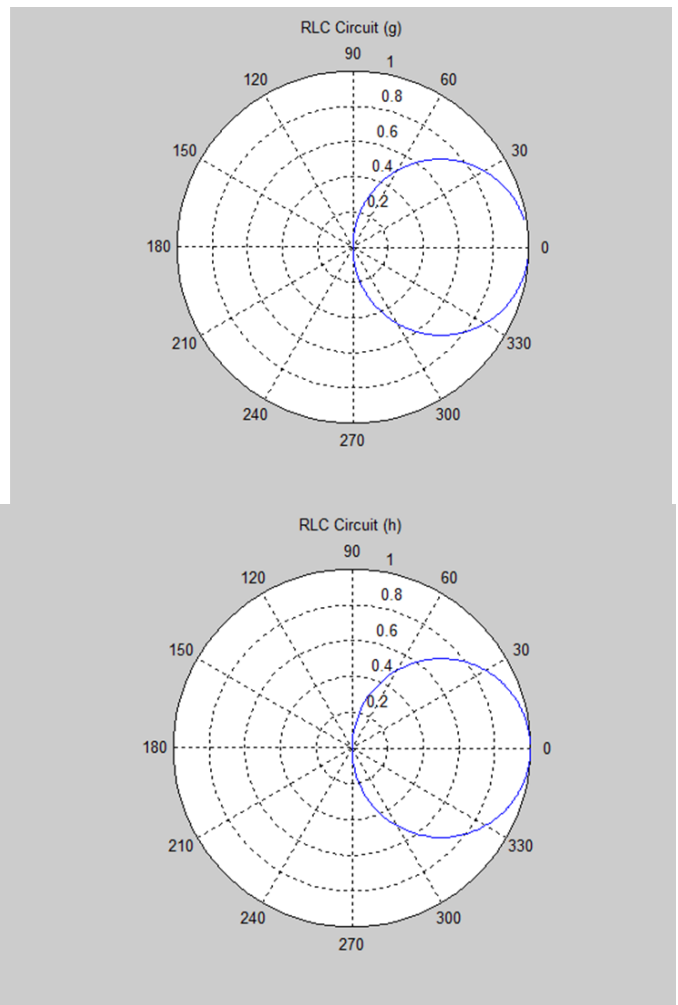
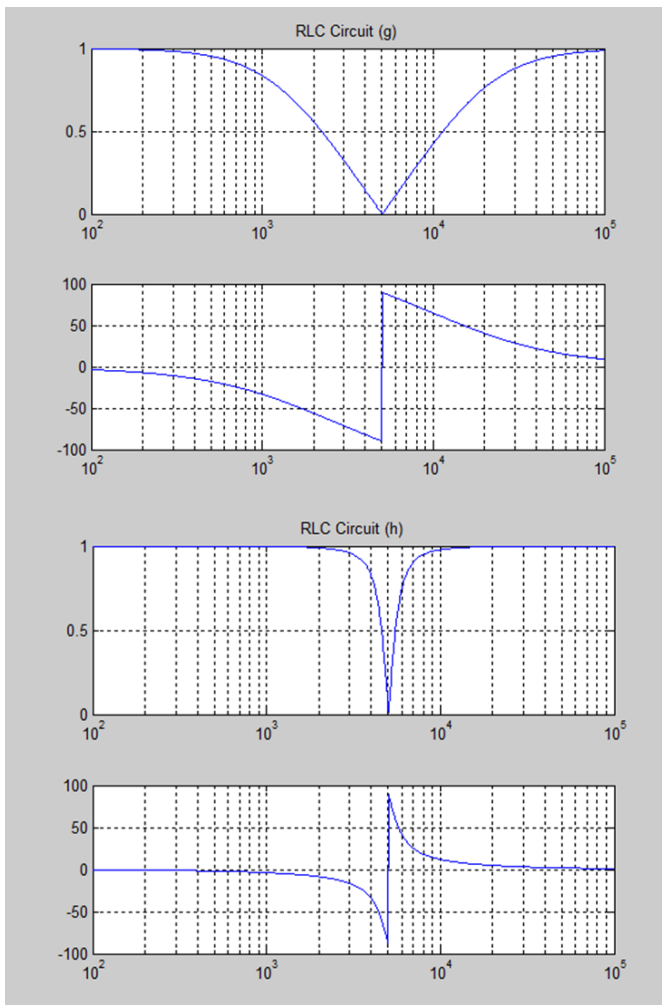
(h) Notch filter: $R=100\Omega$, $L=1\text{mH}$, $C=1\mu\text{F}$.



Outputs







You need to submit a published pdf file that shows these graphs and a .m matlab code electronically to my email: nehmetallah@cua.edu.

Note: You cannot work with other students and share code with other students.